PHYS 143 – Mechanics and Thermal Physics

Motion

How do we describe where an object is, how it moves, and how its motion changes?

Hope Donglo

Office 24, Department of Physics, University of Ghana

Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh

Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

This week: the central question and the thread

CORE

By the end of this week, you should be able to:

1. Distinguish position, displacement, velocity and acceleration, and represent them with diagrams, graphs and equations.
2. Calculate velocity and acceleration by differentiation, and recover them by integration with initial conditions.
3. Select a kinematic model, and state when it is valid.
4. Solve free-fall, projectile, circular and relative-motion problems.

Central question

How do we describe where an object is, how it moves, and how its motion changes?

The thread of the week

Differentiate to go from position to velocity to acceleration; integrate, with initial conditions, to go back. Choose the simplest model that is valid, and say when it applies.

Two contacts: A DESCRIBE, then start to **MODEL**; **B** finish the **MODEL**, then **APPLY**.

Reading: Young and Freedman, Chapter 2, §2.1–2.6 and Chapter 3, §3.1–3.5. **Labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

Not marked. Answer individually, then check the answers.

1. Give the signs of the x- and y-components of a $20\,\mathrm{m/s}$ vector at 30° above the $-x$ axis.

   150° from $+x$: $x < 0$, $y > 0$ (-17.3, $+10.0$).

2. Differentiate $x = 3t^2 - 4t + 2$ and evaluate the result at $t = 2\,\mathrm{s}$.

   $\mathrm{d}x/\mathrm{d}t = 6t - 4$, so $8\,\mathrm{m/s}$.

3. Integrate $a = 6t$, given $v = 0$ at $t = 0$.

   $v = 3t^2$.

4. Solve $4.9t^2 - 12t - 3 = 0$. Which root can be a time?

   $t = \dfrac{12 \pm \sqrt{144 + 58.8}}{9.8} = 2.68\,\mathrm{s}$ or $-0.23\,\mathrm{s}$; only the positive root.

5. What do the slope and the area under a straight-line graph segment mean?

   Slope = rise/run; area = base $\times$ height (triangle: half).

6. Convert $72\,\mathrm{km/h}$ to $\mathrm{m/s}$.

   $72/3.6 = 20\,\mathrm{m/s}$.

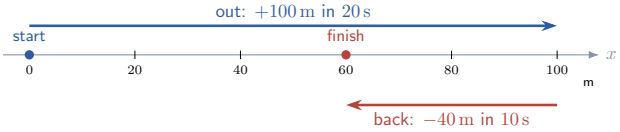
These are the Week 1 and calculus skills this week depends on. A calculus refresher is in the supplementary notes.

Contact A: Describe, then start to Model

How do we represent motion, and how is it changing?

Opening problem

A runner goes $100\,\mathrm{m}$ east in $20\,\mathrm{s}$, turns, and runs $40\,\mathrm{m}$ back west in $10\,\mathrm{s}$. Where is the runner? Over the whole journey, how has the position changed, and how rapidly and in what direction?



Predict first

Before any calculation: is the average velocity positive or negative? Is it larger or smaller than the average speed?

But what if... "how far did the runner go?" has two honest answers, $60\,\mathrm{m}$ from the start or $140\,\mathrm{m}$ of road? We need words that tell them apart: the **vocabulary of motion**.

Defining the vocabulary of motion (1): where, and how far

CORE

Quantity	What it is	What it measures	SI unit	Representation
Position	the vector from the origin to the particle, $\vec{r}$	where the particle is (on a line, the coordinate x)	m	a point on an axis; the vector $\vec{r}$
Displacement	the change in position, $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$	how far and which way the position has changed	m	arrow from start to finish
Distance	total length of the path travelled (a scalar)	how much ground was covered	m	length along the path

Interpret: position answers *where is it?*; displacement answers *how has the position changed?*; distance answers *how much ground was covered?*

But what if... the journey also asks *how rapidly* the position changes? Words are needed for rates of change: **velocity, speed and acceleration**.

Defining the vocabulary of motion (2): how fast, and how it changes

CORE

Quantity	What it is	What it measures	SI unit	Representation
Velocity	the instantaneous rate of change of position (a vector)	how fast and in what direction the particle moves	m/s	tangent to the path; slope of x–t
Speed	distance travelled divided by time; instantaneous speed is the magnitude of the velocity (a scalar)	how fast the particle moves, with no direction	m/s	length of the velocity arrow
Acceleration	the rate of change of velocity with time (a vector)	any change of velocity: speeding up, slowing down or turning	$\mathrm{m/s^2}$	arrow; slope of v–t

Interpret: velocity answers *how fast, and which way?*; acceleration answers *how is the velocity changing?*

But what if... the same word names a vector in one place and a number in another? We must fix the **notation** before using these words in equations.

Notation used in every lecture

Vector: $\vec{v}$ (bold, with an arrow). **Magnitude:** $v = |\vec{v}| \geq 0$, a number with no direction. For velocity, v is the **speed**.

Scalar component: v_x, a *signed number*: the part of $\vec{v}$ along the x axis you chose.

Component vector: $v_x\,\hat{\imath}$. The number v_x is not a vector; $v_x\,\hat{\imath}$ is.

The runner, going west

With $+x$ east:
$v_x = -4.0\,\mathrm{m/s}$ (a signed number)
$\vec{v} = -4.0\,\hat{\imath}\,\mathrm{m/s}$ (a vector)
$v = |\vec{v}| = 4.0\,\mathrm{m/s}$ (the speed)

Careful: choose the axis first

v_x can be negative; the speed v cannot. Choose an origin and positive direction *before* assigning signs; reversing the axis flips every sign. A change is final minus initial.

But what if... the sign of v_x depends on which way we call positive? Then position and displacement need an **origin and an axis** chosen first.

- **Position** $x(t)$: the coordinate that tells us where the object is, relative to a chosen *origin* along a chosen *axis*.
- **Displacement:** the change in the coordinate x,

$$\Delta x = x_2 - x_1 \quad (\text{final} - \text{initial})$$

  It can be positive, negative or zero. It does not depend on the path.
- **Distance travelled:** the total length of the path, a scalar greater than or equal to zero.

Physical meaning

Position depends on the origin and axis you choose. Displacement is independent of the origin, provided the same coordinate system is used for both positions. Its sign depends on which direction you call positive.

Link to Week 1

In 1D, Δx is the x-component of $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$, vector subtraction from last week.

But what if... the runner turns back? $\Delta x = +60\,\mathrm{m}$ but distance $140\,\mathrm{m}$. How rapidly did x change in $30\,\mathrm{s}$? → **average velocity**.

Concept check: distance or displacement?

Which of these can be negative? Which is path-independent?

1. The length of road travelled from Accra to Kumasi.
2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Answers: (1) distance, never negative. (2) displacement, positive here. (3) displacement $= 0$, distance $=$ one lap. (4) distance. Displacement depends only on the end points; distance depends on the whole path.

But what if... we measure x westward instead? Every sign in the runner's journey flips ($\Delta x = -60\,\mathrm{m}$), yet nothing physical changed. Choose and mark the axes first; a change is always final minus initial.

Definitions

The average x-velocity is the x-component of the displacement, Δx, divided by the time interval Δt during which it occurs:

$$v_{\text{av},x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Average speed = distance travelled $/\,\Delta t$.

The signed $v_{\text{av},x}$ can be negative; the average speed cannot.

Opening problem: the runner

A runner goes $100\,\text{m}$ east in $20\,\text{s}$, turns, and runs $40\,\text{m}$ back west in $10\,\text{s}$. Where is the runner? Over the whole journey, how has the position changed, and how rapidly and in what direction?

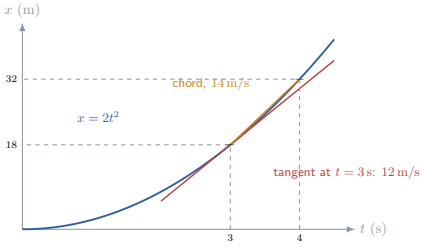
$+x$ east, start at $x = 0$: the runner is at $x = +60\,\text{m}$, $\Delta x = +60\,\text{m}$ east.
$v_{\text{av},x} = +60/30 = +2.0\,\text{m/s}$ east. Distance $140\,\text{m}$, average speed $4.7\,\text{m/s}$.

Evaluate: positive, and smaller than the average speed. The prediction holds.

But what if... one long interval hides what happens inside it? An average over $30\,\text{s}$ cannot give the velocity at one instant. → **average to instantaneous velocity**.

A particle moves with $x = 2t^2$ (SI units).
Shrink the interval after $t = 3\,\mathrm{s}$:

Δt (s)	$v_{\mathrm{av},x}$ (m/s)
1.0	14.0
0.5	13.0
0.1	12.2
0.01	12.02



The chord slope approaches the tangent slope: $12\,\mathrm{m/s}$.

But what if... the table only *approaches* $12\,\mathrm{m/s}$ and we had to guess the limit? Make the limit exact for any t: the **derivative**.

Rule: instantaneous velocity

The instantaneous x-velocity is the limit of the average x-velocity as the time interval approaches zero, the instantaneous rate of change of the x-coordinate:

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}$$ the slope of the tangent to the x–t graph

Reading the graph:

- positive slope $\rightarrow v_x > 0$ (moving in $+x$)
- negative slope $\rightarrow v_x < 0$
- zero slope $\rightarrow$ momentarily at rest
- steeper $\rightarrow$ greater speed $|v_x|$

Two different graphs

An x–t graph is *not* the path of the object. The height of the graph is the position, not the speed.

But what if... we want to be sure we can use these two velocities without any algebra? Test the ideas first: **a concept check**.

Decide, and sketch an x–t graph where it helps

1. Can the average speed over a journey be smaller than $|v_{\text{av},x}|$?
2. Can the velocity be zero while the object is not at the origin?
3. Can the velocity be zero at one instant and the object still be moving a moment later?
4. Can the speed be constant while the velocity changes?

Answers

1. **No.** Distance $\geq |\Delta x|$, so average speed $\geq |v_{\text{av},x}|$.
2. **Yes.** Velocity says how position is changing, not where it is.
3. **Yes.** A ball at the top of its flight has $v = 0$ only for an instant.
4. **Yes.** Constant speed on a curve: the direction of $\vec{\boldsymbol{v}}$ changes (circular motion, Contact B).

But what if... the motion is not along one line, such as a ball thrown across a field? Position then needs x, y (and z): a **vector**.

In two or three dimensions the position vector $\vec{r}$ goes from the origin to the point where the particle is:

$$\vec{r}(t) = x(t)\,\hat{\imath} + y(t)\,\hat{\jmath} + z(t)\,\hat{k}$$

Rule: displacement and velocity as vectors

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 \qquad \vec{v}_{\text{av}} = \frac{\Delta\vec{r}}{\Delta t}$$

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\hat{\imath} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{\jmath} + \frac{\mathrm{d}z}{\mathrm{d}t}\hat{k}$$

Physical meaning

The velocity vector is **tangent to the path** and points in the instantaneous direction of motion. $\vec{v}_{\text{av}}$ has the direction of $\Delta\vec{r}$.

Week 1 tools

Components, signs, magnitude $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ and quadrants carry over.

But what if... $\vec{r}(t)$ is given by formulas for $x(t)$, $y(t)$, $z(t)$? Differentiate each component.

Rule: differentiate component by component

$$\textbf{position } \vec{\boldsymbol{r}}(t) \xrightarrow{\mathrm{d}/\mathrm{d}t} \textbf{velocity } \vec{\boldsymbol{v}}(t) \xrightarrow{\mathrm{d}/\mathrm{d}t} \textbf{acceleration } \vec{\boldsymbol{a}}(t)$$

$\frac{\mathrm{d}}{\mathrm{d}t} t^n = n t^{n-1}$, $\frac{\mathrm{d}}{\mathrm{d}t} \cos \omega t = -\omega \sin \omega t$, $\frac{\mathrm{d}}{\mathrm{d}t} \sin \omega t = \omega \cos \omega t$. **Substitute** the time *after* differentiating.

When to use it

The position (or velocity) is given as formulas in t and you need the velocity (or acceleration). The reverse step, integration, comes later in Contact A.

Watch out and check

$\sin$ and $\cos$ take radians; ωt is dimensionless. Every derivative divides by seconds, so the units must change; the direction of $\vec{\boldsymbol{v}}$ must match the path.

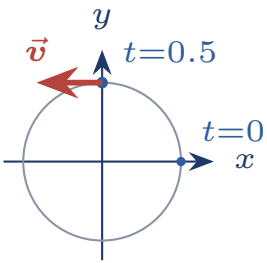
But what if... we try it on a path that curves through space? A circle in the xy-plane combined with a steady rise in z.

Question

A particle has position (SI units) $\vec{\boldsymbol{r}} = 2.0\cos(\pi t)\,\hat{\boldsymbol{\imath}} + 2.0\sin(\pi t)\,\hat{\boldsymbol{\jmath}} + 1.0\,t^2\,\hat{\boldsymbol{k}}$. Find its velocity at $t = 0.5\,\mathrm{s}$.

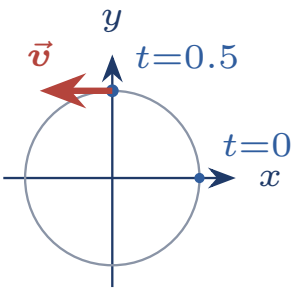
Predict first

At $t = 0$ the particle is at $(2.0, 0, 0)$. In which direction is it moving at $t = 0.5\,\mathrm{s}$, and is z increasing?



Example 1: velocity from position, solution (1): identify and set up

CORE



Key idea

Velocity is the derivative of each component, so each component is handled on its own.

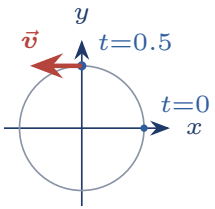
IDENTIFY

Given: $\vec{r}(t)$ above, $t = 0.5\,\mathrm{s}$. Find: $\vec{v}$ and v.

SET UP

Equation: $\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t$, term by term.

Prediction: at the top of the circle, moving toward $-x$, with z increasing.



EXECUTE

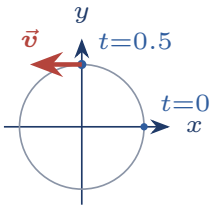
Write the rule. $\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t$: differentiate each component.

Differentiate x. $v_x = \dfrac{\mathrm{d}}{\mathrm{d}t}\left[2.0\cos(\pi t)\right] = -2.0\pi\sin(\pi t)$

Substitute $t = 0.5\,\mathrm{s}$. $v_x = -2.0\pi\sin\frac{\pi}{2} = -2.0\pi(1)$

$$\boxed{v_x = -6.28\,\mathrm{m/s}}$$

Differentiate y. $v_y = \dfrac{\mathrm{d}}{\mathrm{d}t}\left[2.0\sin(\pi t)\right] = 2.0\pi\cos(\pi t)$



EXECUTE, continued

Substitute. $v_y = 2.0\pi \cos\frac{\pi}{2} = 2.0\pi(0)$ $\boxed{v_y = 0}$

Differentiate z, substitute. $v_z = \dfrac{\mathrm{d}}{\mathrm{d}t}\left[1.0t^2\right] = 2.0t = 2.0(0.5)$

$\boxed{v_z = 1.0\,\mathrm{m/s}}$

Speed: write $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$.

$v = \sqrt{(-6.28)^2 + 0^2 + (1.0)^2} = \sqrt{40.5}\,\mathrm{m/s}$ $\boxed{v = 6.36\,\mathrm{m/s}}$

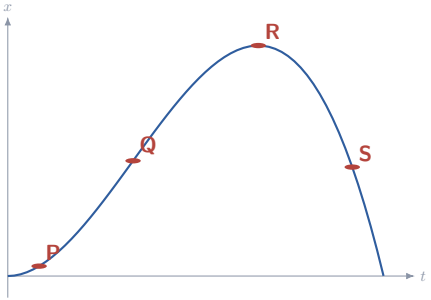
EVALUATE

At $t = 0.5\,\mathrm{s}$ the particle is at $(0, 2.0, 0.25)\,\mathrm{m}$, the top of the circle; $-x$ is counter-clockwise and z increases, as predicted. Units m/s.

But what if... we are given only a *graph* of $x(t)$? Then slopes must be read off the picture.

Question

Of the four points P, Q, R and S: (a) at which is v_x **most positive**? (b) at which is the **speed** greatest? Decide first: which point has the steepest positive slope, which the steepest slope of either sign?



This curve is $x = 2t^2 - t^3/3$, so $v_x = 4t - t^2$:

	t (s)	slope	v_x (m/s)
P	0.5	up	$+1.75$
Q	2	steepest up	$+4.0$
R	4	flat	0
S	5.5	steepest down	-8.25

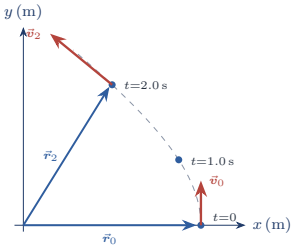
(a) Q, **(b) S**: the speed is $|v_x|$, so a steep *negative* slope counts.

Question

A rover explores the surface of Mars. The stationary lander is the origin and the surface is the xy-plane. The rover's coordinates are (SI units) $x = 2.0 - 0.25\,t^2$ and $y = 1.0\,t + 0.025\,t^3$.
(a) Find its coordinates and distance from the lander at $t = 2.0\,\mathrm{s}$.
(b) Find its displacement and average velocity from $t = 0$ to $2.0\,\mathrm{s}$.
(c) Find its velocity at $t = 2.0\,\mathrm{s}$: components, magnitude and direction.

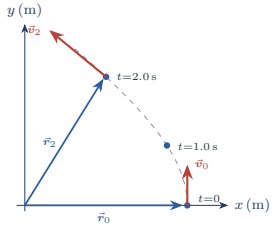
Predict first

From $t = 0$ to $2.0\,\mathrm{s}$, x decreases and y increases. In which quadrant does the average velocity point?



Example 2: a rover on Mars, solution (1): identify and set up

CORE



Key idea

Two positions give the *average* velocity (the chord). The derivative gives the *instantaneous* velocity (the tangent).

IDENTIFY

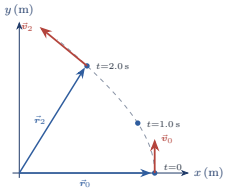
Given: $x(t)$, $y(t)$. Find: $\vec{\boldsymbol{r}}$ and r at $2.0\,\mathrm{s}$; $\Delta\vec{\boldsymbol{r}}$, $\vec{\boldsymbol{v}}_{\mathrm{av}}$; $\vec{\boldsymbol{v}}$, v and direction at $2.0\,\mathrm{s}$.

SET UP

$\vec{\boldsymbol{r}} = x\,\hat{\boldsymbol{\imath}} + y\,\hat{\boldsymbol{\jmath}}$, $\quad \Delta\vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}_2 - \vec{\boldsymbol{r}}_0$, $\quad \vec{\boldsymbol{v}}_{\mathrm{av}} = \Delta\vec{\boldsymbol{r}}/\Delta t$.

$v_x = \mathrm{d}x/\mathrm{d}t$, $\quad v_y = \mathrm{d}y/\mathrm{d}t$. *Reuse:* reconstruct from components (Week 1).

Prediction: the average velocity points up and to the left (second quadrant).



EXECUTE (a)

Put $t = 2.0\,\text{s}$ in x. $x = 2.0 - 0.25(2.0)^2 = 2.0 - 1.0 = 1.0\,\text{m}$

Put $t = 2.0\,\text{s}$ in y. $y = 1.0(2.0) + 0.025(2.0)^3 = 2.0 + 0.20 = 2.2\,\text{m}$

Distance: write $r = \sqrt{x^2 + y^2}$. $r = \sqrt{(1.0)^2 + (2.2)^2} = \sqrt{5.84}$

$$\boxed{r = 2.4\,\text{m}}$$

EXECUTE (b)

Initial position, $t = 0$. $x_0 = 2.0\,\text{m}$, $y_0 = 0$

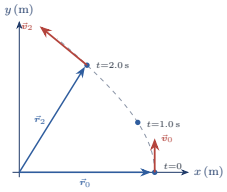
Displacement: write $\Delta\vec{r} = \vec{r}_2 - \vec{r}_0$.

$$\Delta\vec{r} = (1.0 - 2.0)\,\hat{\boldsymbol{\imath}} + (2.2 - 0)\,\hat{\boldsymbol{\jmath}}$$

$$\boxed{\Delta\vec{r} = (-1.0\,\hat{\boldsymbol{\imath}} + 2.2\,\hat{\boldsymbol{\jmath}})\,\text{m}}$$

Average velocity: write $\Delta\vec{r}/\Delta t$. $\vec{v}_{\text{av}} = \dfrac{(-1.0\,\hat{\boldsymbol{\imath}} + 2.2\,\hat{\boldsymbol{\jmath}})\,\text{m}}{2.0\,\text{s}}$

$$\boxed{\vec{v}_{\text{av}} = (-0.50\,\hat{\boldsymbol{\imath}} + 1.1\,\hat{\boldsymbol{\jmath}})\,\text{m/s}}$$



EXECUTE (c)

Differentiate x. $v_x = \dfrac{\mathrm{d}}{\mathrm{d}t}\left[2.0 - 0.25\,t^2\right] = -0.50\,t$

Differentiate y. $v_y = \dfrac{\mathrm{d}}{\mathrm{d}t}\left[1.0\,t + 0.025\,t^3\right] = 1.0 + 0.075\,t^2$

Put $t = 2.0\,\mathrm{s}$ in each. $v_x = -0.50(2.0) = -1.0\,\mathrm{m/s}$,
$v_y = 1.0 + 0.075(2.0)^2 = 1.3\,\mathrm{m/s}$

Magnitude: write $v = \sqrt{v_x^2 + v_y^2}$.
$v = \sqrt{(-1.0)^2 + (1.3)^2} = \sqrt{2.69}$

$$\boxed{v = 1.6\,\mathrm{m/s}}$$

Reference angle: write $\tan\alpha = |v_y|/|v_x|$. $\alpha = \arctan\dfrac{1.3}{1.0} = 52^\circ$ above the $-x$ axis

Direction from $+x$. $180^\circ - 52^\circ$ $\quad\boxed{\theta = 128^\circ}$

EVALUATE

$\vec{v}_{\text{av}}$ points up and left, as predicted. $\vec{v}$ at $2.0\,\mathrm{s}$ points the same way and is faster ($1.6 > |\vec{v}_{\text{av}}| = 1.2\,\mathrm{m/s}$) because v_y grows with t. $\vec{v}$ is tangent to the path.

Common mistakes and the correct habit

Mistake	Instead
Reading an x–t graph as the path of the object.	The height is the position; the slope is the velocity.
Confusing distance with displacement, speed with velocity.	Displacement and velocity are signed; distance and speed are not.
Writing v when a signed component v_x is meant.	$v = \lvert\vec{v}\rvert \geq 0$ is the speed; v_x may be negative.
Assigning signs before choosing the axes.	Choose the origin and the positive direction first, and mark them.
Using the average velocity as the instantaneous velocity.	Instantaneous velocity is the slope of the tangent, not of the chord.
Subtracting initial from final the wrong way round.	A change is always final minus initial.
Thinking a negative v_x or a negative a_x means slowing down.	Compare the two signs: the same sign means speeding up.

From describing motion to modelling how it changes

The runner *turns round*: v_x goes from $+5.0$ to $-4.0\,\mathrm{m/s}$. **How is the velocity itself changing, and can we use that to predict the motion?**

This contact: **acceleration**; **integration**; the **constant-acceleration equations** and free fall.

Poll

Can an object have a **negative** velocity and still be **speeding up**? A car moves toward $-x$ and the driver presses the accelerator. Is v_x negative? Is it speeding up? Is a_x positive or negative?

Commit to an answer now. We settle it with the sign rule, and return to the question in full after it.

But what if... we try to say this without a definition of a_x itself? First we need the rate of change of velocity: **average acceleration**.

Definition

$$\vec{a}_{\text{av}} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} \qquad \text{unit: } \mathrm{m/s^2}$$

In 1D: $a_{\text{av},x} = (v_{2x} - v_{1x})/\Delta t$.
$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1 = \vec{v}_2 + (-\vec{v}_1)$: the same vector subtraction as in Week 1.

Physical meaning

Acceleration is the rate of change of velocity with time: any velocity change, speeding up, slowing down or turning.

Back to the runner

The turn takes $2.0\,\mathrm{s}$ and v_x goes from $+5.0$ to $-4.0\,\mathrm{m/s}$:
$a_{\text{av},x} = \dfrac{-4.0 - 5.0}{2.0} = -4.5\,\mathrm{m/s^2}$.

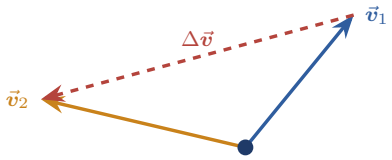
But what if... the velocity changes in both components, as for a jet that turns? Apply the definition to a plane in two dimensions.

Question

A jet plane flies at constant altitude. At $t_1 = 0$ its velocity components are $v_x = 90\,\mathrm{m/s}$, $v_y = 110\,\mathrm{m/s}$. At $t_2 = 30.0\,\mathrm{s}$ they are $v_x = -170\,\mathrm{m/s}$, $v_y = 40\,\mathrm{m/s}$.
(a) Sketch $\vec{v}_1$ and $\vec{v}_2$. How do they differ?
(b) Find the components of the average acceleration.
(c) Find its magnitude and direction.

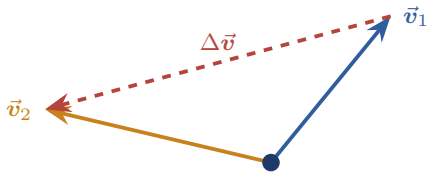
Predict first

Does $\vec{a}_{\text{av}}$ point along $\vec{v}_1$, along $\vec{v}_2$, or along neither?



Example 3: a jet plane, solution (1): identify and set up

CORE



Key idea

$\vec{a}_{\text{av}}$ points along $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$, the vector from the tip of $\vec{v}_1$ to the tip of $\vec{v}_2$. It need not point along either velocity.

IDENTIFY

Given: $\vec{v}_1 = (90, 110)\,\text{m/s}$ at $t_1 = 0$; $\vec{v}_2 = (-170, 40)\,\text{m/s}$ at $t_2 = 30.0\,\text{s}$. Find: components, magnitude and direction of $\vec{a}_{\text{av}}$.

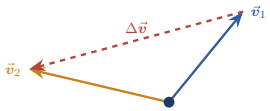
SET UP

$\vec{a}_{\text{av}} = \Delta\vec{v}/\Delta t = (\vec{v}_2 - \vec{v}_1)/(t_2 - t_1)$, component by component. *Reuse:* reconstruct from components.

Prediction: $\Delta\vec{v}$ points left and down, so $\vec{a}_{\text{av}}$ points along neither velocity.

Example 3: a jet plane, solution (2): execute and evaluate

CORE



EXECUTE

Write the definition. $a_{\text{av},x} = (v_{2x} - v_{1x})/(t_2 - t_1)$, $a_{\text{av},y} = (v_{2y} - v_{1y})/(t_2 - t_1)$

Substitute x. $a_{\text{av},x} = (-170 - 90)/30.0$ $\boxed{a_{\text{av},x} = -8.67\,\text{m/s}^2}$

Substitute y. $a_{\text{av},y} = (40 - 110)/30.0$ $\boxed{a_{\text{av},y} = -2.33\,\text{m/s}^2}$

Magnitude: write $a = \sqrt{a_x^2 + a_y^2}$.

$a = \sqrt{(-8.67)^2 + (-2.33)^2} = \sqrt{80.6}$ $\boxed{8.98\,\text{m/s}^2}$

Direction: both negative, third quadrant.

$\theta = 180^\circ + \arctan(2.33/8.67) = 180^\circ + 15.1^\circ$ $\boxed{195^\circ}$

EVALUATE

Both components are negative, so $\vec{a}_{\text{av}}$ points left and down, along neither velocity, as predicted. Check: $|\Delta\vec{v}| = \sqrt{260^2 + 70^2} = 269\,\text{m/s}$ and $269/30.0 = 8.98\,\text{m/s}^2$. ✓

But what if... the runner's $a_x = -4.5\,\text{m/s}^2$ is an *average* over $2.0\,\text{s}$? It cannot say what happens at each instant of the turn. → **instantaneous acceleration**.

Rule: instantaneous acceleration

The instantaneous x-acceleration is the limit of the average x-acceleration as the time interval approaches zero, the instantaneous rate of change of the x-velocity:

$$\vec{\boldsymbol{a}} = \lim_{\Delta t \to 0} \frac{\Delta \vec{\boldsymbol{v}}}{\Delta t} = \frac{\mathrm{d}\vec{\boldsymbol{v}}}{\mathrm{d}t} = \frac{\mathrm{d}^2 \vec{\boldsymbol{r}}}{\mathrm{d}t^2} \qquad a_x = \frac{\mathrm{d}v_x}{\mathrm{d}t} = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}$$

- On a v_x–t graph, a_x is the *slope*.
- On an x–t graph, a_x is the *curvature*: concave up $\to a_x > 0$; concave down $\to a_x < 0$.

The chain:

$$x(t) \xrightarrow{\mathrm{d}/\mathrm{d}t} v_x(t) \xrightarrow{\mathrm{d}/\mathrm{d}t} a_x(t)$$

But what if... $a_x < 0$ throughout the turn, yet the runner first slows and then speeds up? The sign of a_x alone does not say which. $\to$ **compare the signs of v_x and a_x.**

Rule: speeding up or slowing down

Rule: when does speed increase along a line?

Same sign of v_x and a_x $\Rightarrow$ speeding up. **Opposite signs** $\Rightarrow$ slowing down.

v_x	a_x	motion
$+$	$+$	speeding up (moving toward $+x$)
$+$	$-$	slowing down
$-$	$-$	speeding up (moving toward $-x$)
$-$	$+$	slowing down

Common error

A negative acceleration does not by itself mean slowing down, and a negative velocity does not mean slowing down. Compare the two signs.

But what if... we return to the opening poll? Use the rule to answer it for the car driven toward $-x$.

Opening poll

Can an object have a **negative** velocity and still be **speeding up**? A car moves in the $-x$ direction and the driver presses the accelerator. Is its velocity negative? Is it speeding up? Is its acceleration positive or negative?

Answer

Yes. The car moves toward $-x$, so $v_x < 0$. It speeds up, so a_x has the *same* sign as v_x: $a_x < 0$. Negative velocity only says the motion is toward $-x$; whether the speed rises depends on the sign of a_x relative to v_x.

But what if... the velocity changes direction as well as size? The acceleration then has a part *along* the path and a part *across* it.

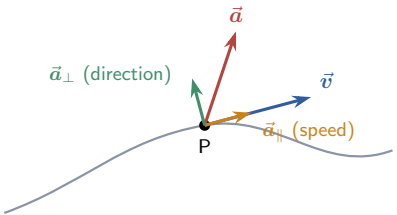
Acceleration along and across the path

Rule: velocity and the path

$\vec{v}$ is tangent to the path.

$\vec{a}_{\parallel}$ (parallel to $\vec{v}$) changes the **speed**.

$\vec{a}_{\perp}$ (perpendicular to $\vec{v}$) changes the **direction**.



A skier who speeds up on a straight slope has only $a_{\parallel}$. A car taking a bend at constant speed has only $a_{\perp}$ (Contact B).

But what if... we are handed a v_x–t graph with straight segments instead of a formula? Slopes give a_x, but how do we recover the displacement? → **slope and area**.

Rule: what slope and area give

Graph	slope gives	area gives
x–t	velocity v_x	(not used)
v_x–t	acceleration a_x	displacement Δx
a_x–t	rate of change of a_x	change in velocity Δv_x

Procedure

Translate both ways: graph $\rightarrow$ description of the motion, and description $\rightarrow$ sketched graph. Read a slope as the vertical change divided by the horizontal change, with units. Areas above the axis add; areas below subtract from the displacement but still add to the distance.

Watch out and check

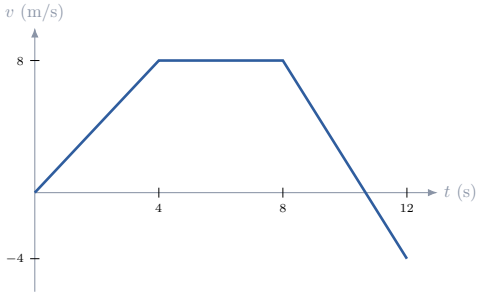
The graph is not the path: an upward graph does not mean the object moves upward. Where $v_x = 0$ the object may reverse, so split the distance calculation there. Do the signs and units of the answer match the graph?

Independent problem

For the motion on the right, find:
(a) the acceleration in each interval and sketch the a_x–t graph;
(b) the displacement from 0 to $12\,\mathrm{s}$;
(c) the distance travelled.

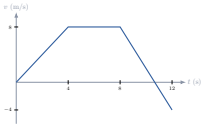
Predict first

Is the displacement larger or smaller than the distance? When does the object reverse?



Your turn: solution, slopes and areas

PRACTICE



IDENTIFY

Given: the v_x–t graph. Find: a_x, displacement, distance.

SET UP

$a_x = \Delta v_x/\Delta t$ (slope); $\Delta x = \int v_x\,dt$ (signed area); split at $v_x = 0$ for (c).

EXECUTE

(a) Slopes:

0–4 s: $a_x = +8/4 = +2.0\,\mathrm{m/s^2}$

4–8 s: $a_x = 0$

8–12 s: $a_x = -12/4 = -3.0\,\mathrm{m/s^2}$

The a_x–t graph is three horizontal steps: $+2$, 0, -3.

(b), (c) Areas:

0–4: $\frac{1}{2}(4)(8) = +16\,\mathrm{m}$

4–8: $4 \times 8 = +32\,\mathrm{m}$

8–10.67: $+10.7\,\mathrm{m}$ (until $v_x = 0$)

10.67–12: $-2.7\,\mathrm{m}$

EVALUATE

Displacement $= 16 + 32 + 10.7 - 2.7 = +\mathbf{56}\,\mathrm{m}$; distance $= 16 + 32 + 10.7 + 2.7 = \mathbf{61.3}\,\mathrm{m}$; distance $\geq$ displacement, as predicted. Reverses at $t = 10.67\,\mathrm{s}$ ($v_x = 0$); area below the axis is negative displacement but positive distance.

But what if... the acceleration is given as a function of time instead of a graph? The area under a_x–t recovers v_x, and the area under v_x–t recovers x. → **integrate, with initial conditions**.

Rule: from acceleration back to position

$$v_x(t) = v_{0x} + \int_0^t a_x \, \mathrm{d}t' \qquad x(t) = x_0 + \int_0^t v_x \, \mathrm{d}t'$$

In vector form: $\vec{\boldsymbol{v}} = \vec{\boldsymbol{v}}_0 + \int_{t_0}^t \vec{\boldsymbol{a}} \, \mathrm{d}t'$ and $\vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}_0 + \int_{t_0}^t \vec{\boldsymbol{v}} \, \mathrm{d}t'$, one component at a time.

Add the initial conditions v_{0x} and x_0; without them the integral is defined only up to a constant. These integrals are the areas under a_x–t (giving Δv_x) and v_x–t (giving Δx).

When to use it

The acceleration (or velocity) is given as a function of time and you need the velocity and position.

Watch out and check

Do not forget the initial value. Check $v(0) = v_0$ and $x(0) = x_0$; differentiate your answer to recover the $a(t)$ you started with.

Contact A: what you can now do

- Tell position, displacement, distance, velocity and speed apart, and find $\vec{\boldsymbol{v}} = \mathrm{d}\vec{\boldsymbol{r}}/\mathrm{d}t$ by differentiating each component.
- Find $\vec{\boldsymbol{a}} = \mathrm{d}\vec{\boldsymbol{v}}/\mathrm{d}t$; v_x and a_x with the same sign mean speeding up.
- Link x–t, v–t and a–t graphs by slope and area, and integrate with initial conditions.

Predict first

Before Wednesday, predict: (1) A ball is thrown straight up. At the top $v_y = 0$. Is a_y zero there? (2) If a_x is constant, must we integrate every time, or is there a shortcut?

But what if... a_x is constant and we want a speed or a position without repeating the integral? → **the constant-acceleration equations, and when they fail**.

Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh

Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

Contact B: Model, then Apply

How do we use the model correctly,
and apply it to complex motion?

Predict first

Predict first. Answers appear one at a time.

1. A ball is thrown straight up. At the top $v_y = 0$. Is a_y zero there? No: v_y is changing as it passes through zero, so $a_y \neq 0$. We confirm this today in free fall.
2. A car moves toward $-x$ and the driver presses the accelerator. What is the sign of a_x? Negative: the same sign as v_x means speeding up.
3. What does the area under an a_x–t graph give? The change in velocity, Δv_x; add v_{0x} to get v_x.

Today's job

Finish the model: when the constant-acceleration equations hold and when they fail. Then use it on motion that is not along a line.

Example 4: vector integration

Question

A particle starts from rest. Its acceleration is $\vec{\boldsymbol{a}} = kt\,(t^2 - 2)\,\hat{\boldsymbol{\imath}} - kt^3\,\hat{\boldsymbol{\jmath}}$, where k is a constant. Find its velocity $\vec{\boldsymbol{v}}(t)$.

Predict first

For small t (with $t^2 < 2$) which way is $\vec{\boldsymbol{a}}$? Which way does the particle start to move?

Units

a is in $\mathrm{m/s^2}$ and t^3 in $\mathrm{s^3}$, so k has units $\mathrm{m/s^5}$.

IDENTIFY

Given: $\vec{\boldsymbol{a}}(t) = kt\,(t^2 - 2)\,\hat{\boldsymbol{\imath}} - kt^3\,\hat{\boldsymbol{\jmath}}$; at $t_0 = 0$ the particle is at rest, $\vec{\boldsymbol{v}}_0 = \vec{\mathbf{0}}$. Find: $\vec{\boldsymbol{v}}(t)$.

SET UP

Vector rule: $\vec{\boldsymbol{v}} = \vec{\boldsymbol{v}}_0 + \int_{t_0}^{t} \vec{\boldsymbol{a}}\,\mathrm{d}t'$. Integrate each component on its own:

$v_x = v_{0x} + \int_0^t a_x\,\mathrm{d}t'$, $\quad v_y = v_{0y} + \int_0^t a_y\,\mathrm{d}t'$. *Reuse:* the one-dimensional integral, with initial conditions.

Prediction: for small t, $a_x < 0$ and $a_y < 0$, so the particle starts to move left and down.

Key idea

Integrating a vector means integrating each component. The initial velocity is also a vector: here it is zero.

EXECUTE

Components of $\vec{a}$. $a_x = k(t^3 - 2t), \quad a_y = -kt^3$

Write the rule for x. $v_x = 0 + \int_0^t k\,(t'^3 - 2t')\,\mathrm{d}t'$

Integrate term by term. $v_x = k\left[\frac{t'^4}{4} - t'^2\right]_0^t$ $\boxed{v_x = k\left(\frac{t^4}{4} - t^2\right)}$

Write the rule for y, integrate. $v_y = 0 - k\int_0^t t'^3\,\mathrm{d}t' = -k\left[\frac{t'^4}{4}\right]_0^t$ $\boxed{v_y = -\frac{kt^4}{4}}$

Result as a vector. $\boxed{\vec{\boldsymbol{v}}(t) = k\left(\frac{t^4}{4} - t^2\right)\hat{\boldsymbol{\imath}} - \frac{kt^4}{4}\,\hat{\boldsymbol{\jmath}}}$

EVALUATE

$\vec{\boldsymbol{v}}(0) = \vec{\mathbf{0}}$. ✓ Differentiating gives $k(t^3 - 2t)$ and $-kt^3$, the given $\vec{\boldsymbol{a}}$. ✓

For small t, $v_x \approx -kt^2 < 0$ and $v_y < 0$: left and down, as predicted. Units: kt^4 is $\mathrm{m/s}$. ✓

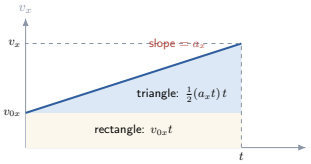
With $k = 1.0\,\mathrm{m/s^5}$ at $t = 2.0\,\mathrm{s}$: $\vec{\boldsymbol{v}} = 1.0(4-4)\,\hat{\boldsymbol{\imath}} - \frac{1.0(16)}{4}\,\hat{\boldsymbol{\jmath}} = -4.0\,\hat{\boldsymbol{\jmath}}\,\mathrm{m/s}$.

But what if... a_x is *constant*? Must we integrate every time, or is there a shortcut? → **the constant-acceleration equations, and when they fail**.

Rule: valid only for constant acceleration along a straight line

(1)	$v_x = v_{0x} + a_x t$	no $x - x_0$
(2)	$x - x_0 = v_{0x} t + \frac{1}{2} a_x t^2$	no v_x
(3)	$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	no t
(4)	$x - x_0 = \frac{1}{2}(v_{0x} + v_x)\, t$	no a_x
(5)	$x - x_0 = v_x t - \frac{1}{2} a_x t^2$	no v_{0x}

Five variables appear: $x - x_0$, v_{0x}, v_x, a_x and t. Each equation omits exactly one. **To choose:** confirm a_x is constant and the motion is on a line; list the knowns and the target; find the variable that is *neither known nor wanted*; use the equation that omits it; solve algebraically, then substitute with units.



If a_x is constant, v_x is a straight line on a v_x–t graph, and the displacement is the area under it: $v_{0x} t$ (rectangle) + $\frac{1}{2} a_x t^2$ (triangle). Eliminating t gives (3); the trapezoid gives (4). Derivations are in the supplementary notes.

But what if... we apply these to the previous example, where $a_x = 6.0t - 4.0$ is not constant? Test it with numbers.

Apply the constant-acceleration equations to the changing-acceleration example

There $a_x = 6.0t - 4.0$ varies. At $t = 3.0\,\text{s}$, $a_x = 14\,\text{m/s}^2$. Suppose we wrongly used that value as if it were constant:

Wrong (constant $a_x = 14$):
$v_x = -3.0 + 14(3.0) = +39\,\text{m/s}$
$x = 2.0 + (-3.0)(3.0) + \frac{1}{2}(14)(9.0) = 56\,\text{m}$

Correct (integration):
$v_x = +12\,\text{m/s}$
$x = 2.0\,\text{m}$

Lesson

Before using equations (1)–(5), ask: *is a_x constant?* If not, integrate.

But what if... a_x *is* constant, and there are five equations to choose from? Pick the one that omits the variable you neither know nor want.

Question

A trotro moving at $25.0\,\mathrm{m/s}$ ($90\,\mathrm{km/h}$) brakes with a constant acceleration of magnitude $6.0\,\mathrm{m/s^2}$ until it stops. How far does it travel, and how long does it take?

eg. a trotro on the Accra–Kumasi road

Predict first

Sketch with $+x$ along the motion. Sign of a_x? A rough distance if the average speed is about half the initial speed and the stop takes about $4\,\mathrm{s}$?



Example 5: solution (1): identify and set up

CORE



Key idea

$x - x_0 \propto v_{0x}^2$: doubling the speed quadruples the braking distance.

IDENTIFY

Given: $v_{0x} = 25.0\,\mathrm{m/s}$, $v_x = 0$, $a_x = -6.0\,\mathrm{m/s^2}$.
Find: distance d and time t.

SET UP

$+x$ along the motion. Choose (3): $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$, which has no t, for d.

Choose (1): $v_x = v_{0x} + a_x t$, which has no $x - x_0$, for t.

Prediction: about $50\,\mathrm{m}$ in about $4\,\mathrm{s}$.

Example 5: solution (2): execute



EXECUTE

Distance: write (3). $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

Solve for $x - x_0$. $x - x_0 = \dfrac{v_x^2 - v_{0x}^2}{2a_x}$

Substitute with units. $x - x_0 = \dfrac{0 - (25.0\,\mathrm{m/s})^2}{2(-6.0\,\mathrm{m/s^2})} = \dfrac{-625}{-12.0}\,\mathrm{m}$ $\boxed{d = 52\,\mathrm{m}}$



EXECUTE, continued

Time: write (1) and solve for t. $t = \dfrac{v_x - v_{0x}}{a_x}$

Substitute with units. $t = \dfrac{0 - 25.0\,\mathrm{m/s}}{-6.0\,\mathrm{m/s^2}}$ $\boxed{t = 4.2\,\mathrm{s}}$

EVALUATE

Average speed $12.5\,\mathrm{m/s}$ times $4.17\,\mathrm{s}$ gives $52\,\mathrm{m}$, matching (4); units m and s; agrees with the estimate. ✓

But what if... two bodies move with different accelerations, such as a motorcycle chasing a car? Write one position equation for each and equate them.

Rule: free fall

If the effects of the air can be ignored, all objects at a location fall with the same downward acceleration, whatever their size or weight. This idealized motion, rising as well as falling, is **free fall**; its constant acceleration has magnitude g:

$$a_y = -g = -9.80\,\mathrm{m/s^2} \quad (+y \text{ upward})$$

It holds **throughout**: rising, at the top, and falling.

Predict first

Hammer and feather dropped in vacuum: which lands first? Neither; both have $a_y = -g$.

Misconception

"At the top the acceleration is zero." It is not: $v_y = 0$ momentarily, but $a_y = -g$ still.

Physical meaning

g is the *magnitude*; the sign of a_y comes from the axis you chose . With $+y$ downward, $a_y = +g$.

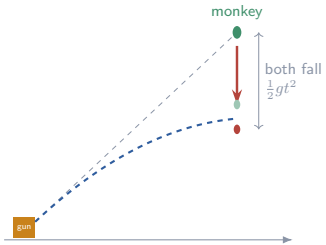
But what if... the motion is not along a line? → **independent components**.

A new question: motion that is not along a line

We have modelled motion along a line. **What changes when the motion is not along a line, or when the observer is moving?** Projectiles, a car on a bend, a walker on a train. **No new kinematics is needed:** this contact reuses the equations of Contact A, component by component. Start with a puzzle.

The monkey and the dart

A dart is aimed directly at a monkey hanging in a tree. When the dart leaves the gun, the monkey lets go and falls. Does the dart hit, pass above, or pass below the monkey? (Assume the dart reaches the monkey's line before the monkey reaches the ground.) Commit to an answer: which of the two experiences gravity?



But what if... the dart moves forward *and* falls at once? We need to know how the horizontal and vertical motions combine.

Rule: independence of components

With uniform gravity and air resistance neglected, the horizontal and vertical component equations can be solved independently. They are linked **only through the common time** t.

A projectile is any object that is given an initial velocity and then follows a path determined entirely by gravitational acceleration and air resistance; the path is its **trajectory**. We model it as a particle with constant acceleration, ignoring air resistance and the curvature and rotation of the Earth.

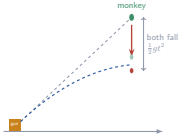
Misconception

No horizontal force acts on a projectile when air resistance is neglected. $a_x = 0$, so v_x is constant.

But what if... we can now settle the monkey and the dart? Apply the rule to the opening puzzle.

Opening puzzle

Selasi aimed dart directly at a monkey hanging in a tree. When the dart leaves the gun, the monkey lets go and falls. Does the dart hit, pass above, or pass below the monkey? (Assume the dart reaches the monkey's line before the monkey reaches the ground.)



Answer: it hits

Without gravity, both the dart and the monkey would stay on the line of sight. With gravity, both fall the *same* distance $\frac{1}{2}gt^2$ below it in time t, because the vertical motion does not depend on the horizontal motion. The dart still meets the monkey.

But what if... we want height, flight time and range as numbers? Write a **model**.

Projectile motion: the model

Assumptions: neglect air resistance; g constant; flat Earth over the range.

Horizontal ($a_x = 0$):

$v_x = v_0 \cos\theta$

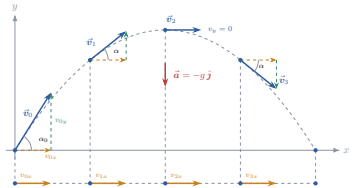
$x = x_0 + v_0 \cos\theta \; t$

Vertical ($a_y = -g$):

$v_y = v_0 \sin\theta - gt$

$y = y_0 + v_0 \sin\theta \; t - \frac{1}{2} g t^2$

At the top: $v_y = 0$ but $a_y = -g$.



Method

Sketch (start, top, end); axes $+x$ horizontal, $+y$ up; resolve $\vec{v}_0$; x-motion with $a_x = 0$; y-motion with $a_y = -g$ *Reuse:* (constant-acceleration equations (1), (2), once per axis); **same time** in both; reconstruct speed and direction if needed. Quadratic in t: keep the physical root.

But what if... we try it on a real launch from level ground? Predict first, then calculate.

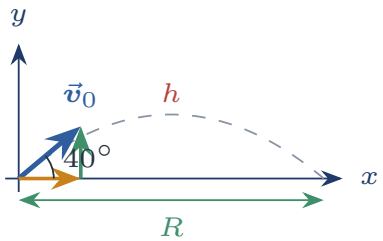
Example 6: a long hit

Question

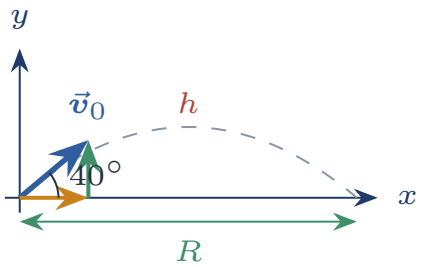
A ball is hit from ground level at $30.0\,\mathrm{m/s}$, 40.0° above the horizontal. Find the time to the highest point, the maximum height, the total time of flight and the horizontal range. (Neglect air resistance; the ground is level.)

Predict first

Sketch. At 45° the range would be $v_0^2/g \approx 92\,\mathrm{m}$; is the range at 40° slightly smaller or larger? What are v_y and a_y at the top?



eg. a batter at the Accra Sports Stadium



Key idea

Horizontal and vertical motions are independent; only the time links them.

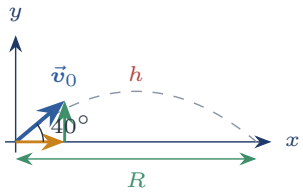
IDENTIFY

Given: $v_0 = 30.0\,\mathrm{m/s}$ at $40.0°$, level ground. Find: time to the top, h, flight time T and range R.

SET UP

Horizontal: $a_x = 0$, $x = v_{0x}t$. Vertical: $a_y = -g$, $v_y = v_{0y} - gt$. *Reuse:* constant-acceleration (1), (2).

Prediction: R slightly less than $92\,\mathrm{m}$; $v_y = 0$ at the top.

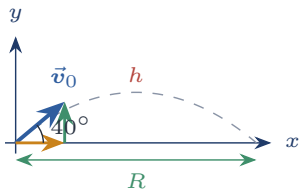


EXECUTE

Resolve $\vec{v}_0$. $v_{0x} = 30.0 \cos 40.0^\circ = 23.0\,\text{m/s}$, $v_{0y} = 30.0 \sin 40.0^\circ = 19.3\,\text{m/s}$

Top: $v_y = 0$. $t = \dfrac{v_{0y}}{g} = \dfrac{19.3}{9.80}$ $\boxed{t = 1.97\,\text{s}}$

Height: (3) with $v_y = 0$. $h = \dfrac{v_{0y}^2}{2g} = \dfrac{(19.3)^2}{2(9.80)}$ $\boxed{h = 19.0\,\text{m}}$



EXECUTE, continued

Landing: $y = 0$. $T = \dfrac{2v_{0y}}{g} = \dfrac{2(19.3)}{9.80}$ $\boxed{T = 3.94\,\text{s}}$ (the other root is $t = 0$)

Range: $x = v_{0x}T$. $R = (23.0\,\text{m/s})(3.94\,\text{s})$ $\boxed{R = 90.4\,\text{m}}$ (unrounded v_{0x}, v_{0y})

EVALUATE

$R < 92\,\text{m}$, as predicted; at the top $\vec{v} = 23.0\,\hat{\imath}\,\text{m/s}$ and $a_y = -9.80\,\text{m/s}^2$; T is twice the time to the top (symmetric path). ✓

But what if... only the *direction* of $\vec{v}$ changes, at constant speed? Does the object accelerate? → **circular motion**.

Poll

A car goes around a circular track at a **constant speed**. Is its acceleration zero?

eg. traffic circles such as Kwame Nkrumah Circle, Accra

Predict first

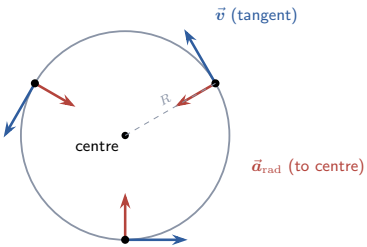
Recall: acceleration is the rate of change of the *velocity vector*. Does the velocity vector stay the same?

No. The speed is constant but the direction of $\vec{v}$ changes continuously, so $\vec{a} \neq 0$. This is $a_\perp$ from Contact A, with no $a_\parallel$.

But what if... $\vec{a} \neq 0$, but in which direction does it point? Draw $\vec{v}$ at two nearby points.

Uniform circular motion: the geometry

CORE



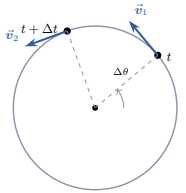
- A circle at constant speed is **uniform circular motion**; $\vec{v}$ is tangent, of constant magnitude v.
- No tangential acceleration: $a_{\text{tan}} = 0$.
- $\vec{a} \perp \vec{v}$, directed toward the centre (never outward): **radial** or **centripetal**.

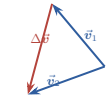
Predict first

If the speed doubles at the same radius, does the acceleration double or quadruple?

But what if... we want its *size*? Compare $\vec{v}$ at two nearby instants.

Why the acceleration is v^2/R





$\Delta\vec{v}$ points toward the centre

In a short time Δt the velocity turns through an angle $\Delta\theta$. The arc is $v\,\Delta t = R\,\Delta\theta$, and $|\Delta\vec{v}| \approx v\,\Delta\theta$ points toward the centre. *Reuse:* $\vec{a} = \Delta\vec{v}/\Delta t$ from Contact A. So

$$a_{\text{rad}} = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{v\,\Delta\theta}{\Delta t} = \frac{v \cdot v\,\Delta t/R}{\Delta t}$$

Rule: radial acceleration

$$a_{\text{rad}} = \frac{v^2}{R}$$

But what if... we are given the number of turns per second, not v? Link v to the period and frequency.

Build the relations from one revolution:

angle 2π rad, distance $2\pi R$

- **Period** T: the time for one revolution (one complete trip around the circle).
- **Frequency** $f = 1/T$ (hertz, $\mathrm{Hz} = \mathrm{s}^{-1}$).
- **Angular speed** $\omega = 2\pi f = 2\pi/T$ (rad/s).
- **Speed:** $v = \dfrac{2\pi R}{T} = \omega R$. *Reuse:* (distance over time, Contact A).

Rule: uniform circular motion

$$a_{\text{rad}} = \frac{v^2}{R} = \omega^2 R = \frac{4\pi^2 R}{T^2}$$

Physical meaning

Constant speed does not mean zero acceleration. For motion on a circle at constant speed, $\vec{a}$ points to the centre and has magnitude v^2/R.

But what if... a passenger walks inside a moving car or train: whose velocity do we mean? → **relative velocity**.

Discussion

A passenger walks along the aisle of a train moving at $20.0\,\mathrm{m/s}$ relative to the ground. How fast is the passenger moving?

Predict first

The passenger walks toward the front at $1.5\,\mathrm{m/s}$ relative to the train. Faster or slower than the train, relative to the ground? What if the passenger walks toward the rear?

The same motion has different velocities for different observers.

But what if... the question has no single answer until we say *relative to what*? We need a notation that records whose velocity, and relative to which observer.

Rule: relative velocity

$$\vec{\boldsymbol{v}}_{P/A} = \vec{\boldsymbol{v}}_{P/B} + \vec{\boldsymbol{v}}_{B/A}$$

Read $\vec{\boldsymbol{v}}_{P/A}$ as "the velocity of P relative to A". Also $\vec{\boldsymbol{v}}_{A/B} = -\vec{\boldsymbol{v}}_{B/A}$. The inner subscript B cancels in the chain: P/B followed by B/A gives P/A. Write the relation *before* inserting values. In 2D, use components and reconstruct (Week 1). *Reuse:* vector addition from Contact 1.

Watch out

Do not add speeds: add *vectors*.

Back to the train: P, T, G

$v_{T/G,x} = +20.0\,\mathrm{m/s}$; toward the front,
$v_{P/T,x} = +1.5\,\mathrm{m/s}$.

$v_{P/G,x} = v_{P/T,x} + v_{T/G,x} = 1.5 + 20.0 = 21.5\,\mathrm{m/s}$.

Toward the rear: $v_{P/T,x} = -1.5\,\mathrm{m/s}$, so $v_{P/G,x} = 18.5\,\mathrm{m/s}$: still forward relative to the ground, slower than the train.

Evaluate: the signs follow from the chosen $+x$.

But what if... the velocities are not along one line, as for a boat crossing a river? Draw the vector triangle.

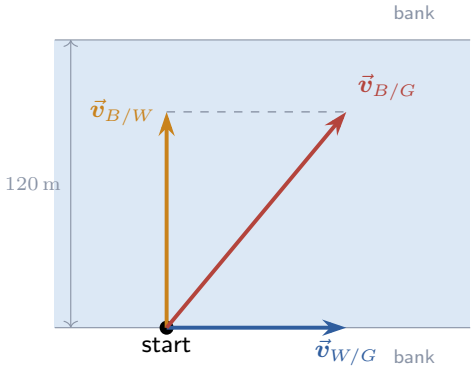
Question

A river flows east at $3.0\,\mathrm{m/s}$ relative to the ground and is $120\,\mathrm{m}$ wide. A boat has speed $4.0\,\mathrm{m/s}$ relative to the water and is pointed due north. Find its velocity relative to the ground, the crossing time and where it lands.

eg. a river such as the Pra (Ghana) or the Nile

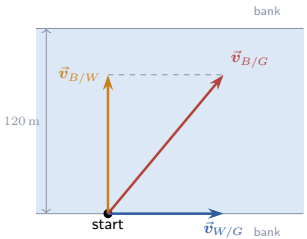
Predict first

Which way is the boat swept? Is the ground speed more or less than $4.0\,\mathrm{m/s}$?



Example 7: solution (1): identify and set up

CORE



Key idea

The current adds sideways motion but does not change the crossing time.

IDENTIFY

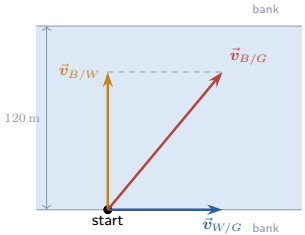
Given: $\vec{v}_{W/G} = 3.0\,\mathrm{m/s}$ east, $\vec{v}_{B/W} = 4.0\,\mathrm{m/s}$ north, width $120\,\mathrm{m}$. Find: $\vec{v}_{B/G}$, crossing time, drift.

SET UP

x east, y north: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$. Time from the northward motion: $t = d/v_y$; drift $= v_x t$.

Reuse: $x = v_x t$, constant velocity.

Prediction: swept east; ground speed $> 4.0\,\mathrm{m/s}$.



EXECUTE

Write the rule. $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$

Substitute the components. $\vec{v}_{B/G} = 4.0\,\hat{j} + 3.0\,\hat{i}$

$$\boxed{\vec{v}_{B/G} = (3.0\,\hat{i} + 4.0\,\hat{j})\ \mathrm{m/s}}$$

Speed. $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(3.0)^2 + (4.0)^2} = \sqrt{25}$ $\boxed{v = 5.0\ \mathrm{m/s}}$

Direction. $\tan\varphi = \dfrac{v_x}{v_y} = \dfrac{3.0}{4.0}$, so $\varphi = 36.9^\circ$ east of north

Time: write $t = d/v_y$**.** $t = \dfrac{120\ \mathrm{m}}{4.0\ \mathrm{m/s}}$ $\boxed{t = 30\ \mathrm{s}}$

Drift: write $v_x t$**.** $(3.0\ \mathrm{m/s})(30\ \mathrm{s})$ $\boxed{90\ \mathrm{m}}$

EVALUATE

Swept east, $5.0 > 4.0\ \mathrm{m/s}$, as predicted;
$\sqrt{90^2 + 120^2} = 150\ \mathrm{m} = 5.0 \times 30$. ✓

But what if... several ideas appear in one problem and nobody names the method? → **integrated application**.

Example 8: dropping a crate (integrated problem)

CORE

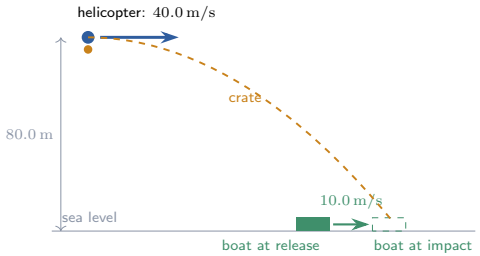
Application problem: decide what physics applies

A helicopter flies horizontally at $40.0\,\mathrm{m/s}$, $80.0\,\mathrm{m}$ above the sea. A boat below moves in the same direction at $10.0\,\mathrm{m/s}$. A crate released from the helicopter is to land on the boat.

(a) How far ahead of the crate (horizontally) must the boat be at release?

(b) With what speed does the crate hit the boat, as seen from the boat?

eg. fishing boats off Elmina, Central Region

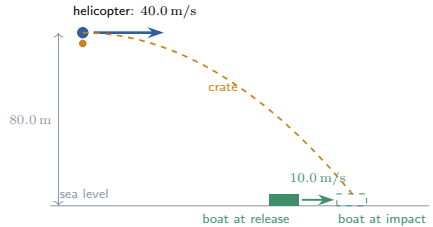


Predict first

IDENTIFY → SET UP (frame? axes? sketch) → PREDICT → EXECUTE → EVALUATE.

Example 8: solution (1): identify and set up

CORE



Key idea

The crate falls for the same time in any frame; only its horizontal speed relative to the boat changes.

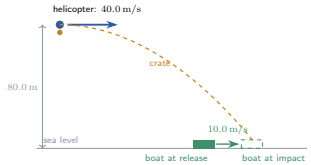
IDENTIFY

Given: $\vec{\boldsymbol{v}}_{C/G} = 40.0\,\hat{\boldsymbol{\imath}}\,\mathrm{m/s}$ at release, $h = 80.0\,\mathrm{m}$, $\vec{\boldsymbol{v}}_{B/G} = 10.0\,\hat{\boldsymbol{\imath}}\,\mathrm{m/s}$. Find: (a) the gap, (b) the speed relative to the boat.

SET UP

$+x$ forward, $+y$ up. Vertical: $y = y_0 - \frac{1}{2}gt^2$, $v_y = -gt$. Relative: $\vec{\boldsymbol{v}}_{C/B} = \vec{\boldsymbol{v}}_{C/G} - \vec{\boldsymbol{v}}_{B/G}$. *Reuse:* free fall, Contact B; relative velocity above.

Prediction: $t \approx 4\,\mathrm{s}$, gap about $120\,\mathrm{m}$.



EXECUTE

Fall time: $y = y_0 - \frac{1}{2}gt^2 = 0$. $t = \sqrt{\dfrac{2y_0}{g}} = \sqrt{\dfrac{2(80.0\,\text{m})}{9.80\,\text{m/s}^2}}$

$\boxed{t = 4.04\,\text{s}}$

Relative horizontal speed.

$v_{C/B,x} = v_{C/G,x} - v_{B/G,x} = 40.0 - 10.0 = 30.0\,\text{m/s}$

(a) Gap: $d = v_{C/B,x}\,t$. $d = (30.0\,\text{m/s})(4.04\,\text{s})$ $\boxed{d = 121\,\text{m}}$

Vertical speed: $v_y = -gt$. $v_{C/B,y} = -(9.80)(4.04) = -39.6\,\text{m/s}$

(b) Speed. $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(30.0)^2 + (-39.6)^2} = \sqrt{2468}$

$\boxed{v = 49.7\,\text{m/s}}$

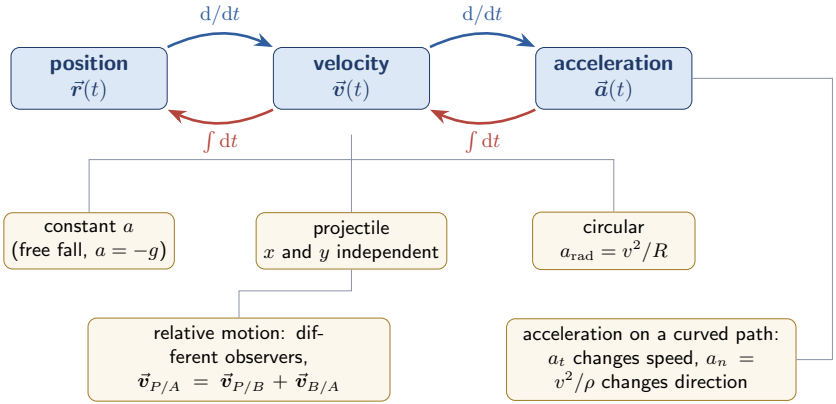
EVALUATE

Units m and m/s; $4.04\,\text{s}$ is plausible for an $80\,\text{m}$ fall ($\sqrt{2h/g}$); relative to the boat the horizontal speed is smaller but the fall time is the same. ✓

Common mistakes and the correct habit

Mistake	Instead
Integrating without the initial conditions.	Add v_{0x} and x_0, then check $v(0)$ and $x(0)$.
Using the constant-acceleration equations when a_x varies.	Ask whether a_x is constant; if not, integrate.
Setting $a_y = 0$ at the top of a flight.	$a_y = -g$ throughout; only v_y is zero there.
Keeping the quadratic root with $t < 0$.	Choose the physical root and say why.
A horizontal force on a projectile.	None is needed: $a_x = 0$ and v_x is constant (no air resistance).
Using different times for x and y.	The common time is the only link between the components.
Setting $a = 0$ at constant speed on a circle.	$a = v^2/R$ toward the centre.
Adding speeds in relative motion.	Add velocities as vectors with the subscript chain.

Week 2 concept map



Key ideas

- **Position is a vector from the origin you choose; displacement $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$ does not depend on the origin; distance travelled is a non-negative scalar.**
- **Velocity $\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t$ is a vector tangent to the path; speed $v = |\vec{v}|$ is its magnitude; v_x is a signed number.**
- Acceleration $\vec{a} = \mathrm{d}\vec{v}/\mathrm{d}t$: the part parallel to $\vec{v}$ changes the speed; the perpendicular part changes the direction.
- Slope and area link the x–t, v–t, a–t graphs. Along a line, v_x and a_x with the same sign mean speeding up.
- **The constant-acceleration equations hold only when a_x is constant along a line; otherwise integrate, with initial conditions.**
- Free fall without air resistance: $a_y = -g$ throughout the flight, including the top.
- **In projectile motion (no air resistance) the horizontal and vertical motions are independent and linked only through the time.**
- Constant speed on a circle still means $a = v^2/R \neq 0$, directed to the centre.
- Velocities add as vectors, with the subscript chain $\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$.
- Habit: sketch, predict, calculate, evaluate.

Reference tools: the next slides (master toolkit, core equations). The Reference deck holds the toolkits and practice.

• MASTER TOOLKIT – For a motion problem

1. IDENTIFY Which object, which interval, what is asked?
2. FRAME Relative to which observer?
3. CHOOSE AXES Origin and positive directions.
4. SKETCH Trajectory or motion diagram; add $\vec{v}$, $\vec{a}$.
5. PREDICT Signs, direction, trend, rough size.
6. SELECT MODEL General $\vec{r}, \vec{v}, \vec{a}$? Constant a? Free fall? Projectile? Circular? Relative?
7. EXECUTE Key move for the model (right); signs and units.
8. EVALUATE Units, sign, direction, graph, assumptions, reasonableness.

If the result disagrees with the prediction, investigate. Seven toolkits: Reference deck.

Step 7: the key move for each model

Formulas in t: differentiate each component; integrate with initial conditions.
Graph: slope gives the rate, area the change.
Constant a: use the equation that omits the variable you neither know nor want.
Free fall: $a_y = -g$ throughout; keep the physical root.
Projectile: resolve; $a_x = 0$, $a_y = -g$; same time.
Circular: $a = v^2/R = \omega^2 R$ to the centre; SI units first.
Relative: $\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$ first; add vectors.

Framework: IDENTIFY (1–2), SET UP (3–6), EXECUTE (7), EVALUATE (8).

Reference tools: core equations

Idea	Equation	Conditions and units
Displacement	$\Delta x = x_2 - x_1, \quad \Delta\vec{r} = \vec{r}_2 - \vec{r}_1$	m; final minus initial
Velocity	$\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t, \quad v_{\mathrm{av},x} = \Delta x/\Delta t$	m/s; slope of x–t
Acceleration	$\vec{a} = \mathrm{d}\vec{v}/\mathrm{d}t = \mathrm{d}^2\vec{r}/\mathrm{d}t^2$	m/s^2; slope of v_x–t
Speeding up	v_x and a_x same sign: speeds up; opposite: slows	motion along a line
Integration	$v_x = v_{0x} + \int a_x \mathrm{d}t, \quad \vec{v} = \vec{v}_0 + \int \vec{a}\,\mathrm{d}t$	each component; needs initial conditions
Constant a_x	$v_x = v_{0x} + a_x t, \quad v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	only if a_x is constant, straight line
Free fall	$a_y = -g, \quad g = 9.80\,\mathrm{m/s^2}$	$+y$ upward; no air resistance
Projectile	$a_x = 0, \quad a_y = -g$; same t	$v_{0x} = v_0\cos\theta$, $v_{0y} = v_0\sin\theta$
Circular motion	$a_{\mathrm{rad}} = v^2/R = \omega^2 R, \quad a_{\tan} = \mathrm{d}v/\mathrm{d}t$	radial toward the centre; $a_{\tan} = 0$ if uniform
Curved path	$a_t = \mathrm{d}v/\mathrm{d}t, \quad a_n = v^2/\rho$	speed and direction changes
Relative velocity	$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$	vector sum; m/s

Where this goes next

Graphs, slope and area

The slope and area of motion graphs return in Lecture 4, where the area under a force–position graph is the work done. Lecture 3 relates the acceleration to the force: $a_x = F_x/m$.

Constant acceleration and free fall

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ becomes the work–energy theorem in Lecture 4; free fall leads to gravitational potential energy in Lecture 5.

Projectiles

Lecture 5 finds the speed at any height by energy conservation, with no time; Lecture 7 treats collisions in two dimensions with the same component method.

Circular motion

Lecture 3 supplies the force behind $a_{\text{rad}} = v^2/R$ (centripetal force); Lecture 8 extends ω, T and f to rotating bodies.

Relative velocity

Lecture 7: a collision looks different in different frames, and the velocity of the centre of mass is the natural frame in which to describe it.

Physical meaning

Independent components, the constant-acceleration equations, integration with initial conditions and $\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t$ remain the core of the motion toolbox for the rest of the course.

Checklist: can I . . .

CORE

- ☐ tell displacement from distance, and velocity from speed, in a given journey?
- ☐ choose an origin and positive direction, and give each sign its meaning?
- ☐ find $\vec{v}$ and $\vec{a}$ from $\vec{r}(t)$ by differentiating each component?
- ☐ find $v_x(t)$ and $x(t)$ from $a_x(t)$ by integrating, with initial conditions?
- ☐ read slopes and areas on x–t, v–t and a–t graphs, and sketch one from a description?
- ☐ decide from the signs of v_x and a_x whether an object speeds up or slows down?
- ☐ choose a constant-acceleration equation, and show that a_x is constant first?
- ☐ solve a free-fall problem with a consistent sign convention, and reject the non-physical root?
- ☐ solve a projectile problem with x and y linked only by time?
- ☐ explain why uniform circular motion has acceleration, and calculate a_{rad}?
- ☐ find a velocity relative to a second observer with the subscript chain?
- ☐ use IDENTIFY → SET UP → EXECUTE → EVALUATE, and choose the model in an unfamiliar problem?

Next

Tutorial sheet: the train and police-car problems and extra graph, free-fall, projectile, circular and relative-velocity problems. The Reference deck holds the supporting material cut from the live lecture; the supplementary notes hold derivations, the calculus refresher and the parabola extension.